International Journal of Dental Hygiene

Official Journal of the International Federation of Dental Hygienists

ORIGINAL ARTICLE

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Periodontal therapy and glycaemic control among individuals with type 2 diabetes: reflections from the PerioCardio study

Abstract: *Objectives:* Diabetes mellitus and periodontal disease are highly prevalent among Indigenous Australian adults. Untreated periodontitis impacts glycaemic control in people with diabetes. The aim of this study was to report on the effect of periodontal therapy on glycaemic control among people with obesity. *Methods:* This subgroup analysis is limited to 62 participants with diabetes from the original 273 Aboriginal Australian adults enrolled into the PerioCardio study. Intervention participants received full-mouth non-surgical periodontal scaling during a single, untimed session while controls were untreated. Endpoints of interest included change in glycated haemoglobin (HbA1c), C-reactive protein (CRP) and periodontal status at 3 months post-intervention. *Results:* There were more females randomized to the treatment group (n = 17) than control (n = 10) while the control group had a higher overall body mass index (BMI) [mean (SD)] 33.1 (9.7 kg m^{-2}) versus 29.9 (6.0 kg m^{-2}). A greater proportion of males were followed up at 3 months compared to females, P = 0.05. Periodontal therapy did not significantly reduce HbA1c: ANCOVA difference in means 0.22 mmol mol^{-1} (95% CI −6.25 to 6.69), CRP: ANCOVA difference in means 0.64 (95% CI −1.08, 2.37) or periodontal status at 3 months. *Conclusions:* Non-surgical periodontal therapy did not significantly reduce glycated haemoglobin in participants with type 2 diabetes. Reasons are likely to be multifactorial and may be influenced by persistent periodontal inflammation at the follow-up appointments. Alternatively, the BMI of study participants may impact glycaemic control via alternative mechanisms involving the interplay between inflammation and adiposity meaning HbA1c may not be amenable to periodontal therapy in these individuals.

Key words: diabetes mellitus, type 2; Indigenous Australian; periodontal therapy, non-surgical; randomized controlled trial

Introduction

Disparities in the health status of Indigenous Australians are well known. To illustrate, 11% of Indigenous Australians have diabetes, 60% of which is poorly controlled (1). The prevalence of type 2 diabetes is more than three times higher than in non-Indigenous Australians (2). Furthermore, it has been estimated that an additional five per cent of Indigenous Australians have impaired fasting glucose metabolism placing them at risk for diabetes (1). In terms of risk factors for other chronic conditions, almost

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Dates:
Accepted 13 May 2016

Clinical trials registration: [ANZCTR number: 12610000817044]. The full trial protocol can be found at: http://www.anzctr.org.au/TrialSearch.aspx using the identification number recorded above.

To cite this article:
Int J Dent Hygiene
DOI: 10.1111/idh.12234
Kapellas K, Mejia G, Bartold PM, Skilton MR, Maple-Brown LJ, Slade GD, O'Dea K, A Brown, Celermajer DS, Jamieson LM. Periodontal therapy and glycaemic control among individuals with type 2 diabetes: reflections from the PerioCardio study.

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half of Indigenous Australians aged 15 years or older are current smokers compared to approximately 20% of the general Australian population(3), 28% are overweight, and a further 29% are obese (4).

Information on the oral health status of Indigenous Australian adults is somewhat limited. Two recently published cross-sectional studies have reported a high prevalence of severe periodontitis is associated with both smoking and diabetes (5, 6). Periodontitis is understood to be a complication of poorly controlled diabetes (7), and conversely, chronic periodontal inflammation may impede glycaemic control leading some to suggest that a bidirectional relationship exists (8).

Systematic reviews published within the past 5 years have reported that periodontal treatment often leads to modest improvements in glycaemic control (9, 10). It is, however, presently unclear whether the provision of periodontal therapy has similar benefits on glycaemic control in Aboriginal Australians, a group that more often express multiple risk factors which when combined may compromise the effectiveness of periodontal interventions.

The PerioCardio study was a parallel-group, randomized, open-label trial investigating the effects of non-surgical periodontal therapy on surrogate markers of cardiovascular disease (11). In response to recent discussion concerning research into periodontal treatment effects on glycaemic control (12, 13), the aim of this study was to evaluate results arising from secondary analysis of the PerioCardio study focusing on change in glycated haemoglobin (HbA1c) in people with diabetes. Findings from this study are extrapolated to discuss implications for future research into periodontal/diabetes relationship in people with obesity.

Study population and methodology

Detailed descriptions of the PerioCardio study methods have been published (11, 14). Briefly, baseline recruitment commenced in June 2010 and completed in January 2012 with the final annual assessment completed in December 2012. Study participants were examined at baseline and again 3 months post-treatment for repeat assessments of periodontal status and glycaemic control. Eligibility criteria were as follows: Aboriginal Australian participants aged 18 years or older without a previous history of cardiovascular disease, a minimum of five natural teeth and moderate/severe periodontitis defined using the joint Centers for Disease Control and Prevention and American Academy of Periodontology case definition(15). Exclusion criteria were as follows: individuals receiving periodontal treatment in the preceding 6 months, those with cardiovascular disease history, rheumatic fever (16) or any other medical condition requiring preventive antibiotic prophylaxis, pregnant women, or people with clinically visible endodontic or orofacial infections.

Participants were recruited from four locations in Australia's Northern Territory. Darwin, the capital and largest urban centre in the Northern Territory, Katherine, a regional town approximately 300 km south of Darwin. Participants were also recruited from two correctional facilities, one in Darwin and the other in Alice Springs (the second largest regional centre in the Northern Territory situated approximately 1,500 km south of Darwin).

Oral assessment method to measure periodontal status

Two dental clinicians, an oral health therapist (examiner #1) and a dentist (examiner #2), conducted the periodontal assessments to measure probing

depth (PPD) and gingival recession which permitted calculation of clinical attachment loss (CAL). Measurements were made at four sites at every tooth excluding third molars using a PCP2 periodontal probe with 2-mm markings. Probing sites included the mesio-buccal, mid-buccal, disto-buccal and disto-lingual. These four sites were selected for convenience and ease of visibility. Gingival bleeding on probing score was collected for each tooth periodontally assessed and was scored based on the Gingival Index criteria (17). Periodontal measurements were standardized to within 1 mm between examiners. Prior to study commencement, five healthy volunteers were examined by both clinicians each blinded to the assessment of the other. Where differences in the assessments occurred, agreement was attained via discussion. Interexaminer reliability between the two clinicians throughout the study was assessed using volunteers not involved in the present trial. Precise agreement on the number of sites with PD ≥2 mm was used as a cut-point with weighted kappa statistic of 0.75 (95% CI 0.70 to 0.80) representing very good agreement (18).

Measurement of blood markers

Non-fasting venous blood samples were collected via the antecubital vein. Samples were transported to a local commercial pathology clinic for analysis of lipid profile: total cholesterol (TC), high-density lipoprotein (HDL) and glycated haemoglobin (HbA1c). HbA1c was determined by turbidimetric inhibition immunoassay with a COBAS INTEGRA (Roche Diagnostics, Indianapolis, USA). Direct methods were used to determine lipid profile using an ADVIA 2400 Chemistry System (Siemens, Tarrytown, USA). Non-HDL cholesterol was the difference between TC and HDL. Following centrifugation, serum and plasma were stored at −80°C until batch analysis. Serum high-sensitivity C-reactive protein level (hsCRP) was measured by particle-enhanced immunonephelometry using the BN II system. Plasma interleukin level (IL-6) was measured via commercial ELISA (Human IL-6 Quantikine kit, R&D Systems Inc., Minneapolis, USA).

Diabetes status was primarily via self-report (n = 41) to an affirmative response to the question '*has a doctor ever told you that you have diabetes?*' Accounting for the possibility that some study participants with diabetes were undiagnosed, diabetes was also confirmed when baseline HbA1c was ≥47.5 mmol mol^{-1} (≥6.5% yielding an additional 21 individuals). These 21 previously undiagnosed study participants were subsequently informed of their diabetes status following presentation of laboratory findings. All 62 study participants had type 2 diabetes mellitus.

Randomization

Participants were randomized on a 1:1 basis to either intervention or control group using permuted block randomization with variable block sizes, stratified by recruitment location. For this purpose, a randomization database was created by a member of the Clinical Trials branch, Baker IDI Heart and Diabetes Institute, Melbourne, who had no other involvement in this study. Randomization was allocated by study clinicians unaware of block sizes following baseline measures by entering the study participant ID and date of measure into the randomization database. Due to the mode of intervention, clinicians and study participants could not be blinded from allocation grouping.

Interventions

The periodontal intervention was performed by the same clinician that conducted the baseline examination and followed baseline measures for the treatment group and was delayed for 12 months in those randomized to the control group. In each instance, the intervention consisted of an untimed single episode of non-surgical periodontal therapy comprising supra- and subgingival scaling using Gracey curettes (Hu-Friedy, Chicago, USA), and piezoelectric ultrasonic device (Kyungwon Ferrite, Gyeonggi-Do, South Korea) with universal tips. To complete the intervention, a full-mouth dental prophylaxis using a rubber cup and Ainsworth prophylaxis paste (Ainsworth Dental Company, Sydney, Australia) was provided to remove plaque and tooth stains. Local anaesthesia was administered as required. All participants received oral hygiene advice consisting of toothbrush and flossing instructions and were provided with a toothbrush and toothpaste at baseline.

Statistical analysis

The extent of PPD ≥4 mm and CAL ≥3 mm was calculated as the total number of sites exhibiting these thresholds divided by the total number of sites examined for each study participant. Calculation of extent scores allows for comparisons of study participants that may have different numbers of teeth. Analyses of change in HbA1c, hsCRP and periodontal status were based on the complete-case approach using analysis of covariance (ANCOVA) for change at 3 months post-baseline with randomization as the factor following adjustment for baseline values as covariates. Post hoc power calculation for change in HbA1c at 3 months was calculated using the *twosamplemeans* feature of PROC POWER in SAS 9.3 for Windows, Cary, N.C., USA.

Ethical approval

The PerioCardio study was approved by the Human Research Ethics Committee of the Northern Territory Department of Health and Menzies School of Health Research, the Central Australian Human Research Ethics Committee, Northern Territory Correctional Services Research Committee, University of Adelaide Human Research Ethics Committee, and the Aboriginal Health Council of South Australia. Study participants gave informed consent before participating which included acknowledgement that should they be randomized into the control group that they would not receive immediate periodontal therapy. Research was conducted in accordance with the World

Medical Association Declaration of Helsinki (version VII, 2008).

Results

Among 62 participants with type 2 diabetes at baseline, there were proportionally more males in the control group compared to treatment and similarly, more current and fewer former smokers in the treatment versus control groups (columns 2 and 3, Table 1). Proportionally, there were more than twice the number of participants with BMI ≥30 kg m^{-2} in the control group relative to treatment. Aside from a higher BMI, there were no other notable differences to age, oral health status, inflammation or blood biochemistry (columns 2 and 3, Table 1).

A significant reduction in the extent of periodontal pockets ≥ 4 mm and Gingival Index was found when comparisons of whole groups from baseline and follow-up were made (columns 4 and 5, Table 1). Comparison for those followed up at 3 months by randomization group revealed that more males were successfully completed, and the Gingival Index was significantly lower in the treatment group: mean (SD) 1.0 (0.5) relative to control 1.5 (0.6), $P = 0.01$. In contrast, IL-6 concentration was significantly higher in the treatment group: mean (SD) 3.7 (2.1) versus control 1.8 (2.5), $P = 0.02$ (columns 7 and 8, Table 1). Other characteristics including age, sex, BMI, smoking status, CRP and metabolic markers did not differ between either group-level and across randomization categories.

Table 1. **Characteristics of study participants with type 2 diabetes at baseline and 3-month follow-up**

	Baseline (*n* = 62)		Group			3 month (*n* = 44)		
Characteristics	Treatment (*n* = 35)	Control (*n* = 27)	Baseline (*n* = 62)	3 month (*n* = 44)	*P*-value	Treatment (*n* = 24)	Control (*n* = 20)	*P*-value
Mean age (SD)	45.5 (10.9)	46.4 (9.1)	45.9 (10.1)	46.9 (10.3)	0.62	48.8 (10.6)	44.6 (9.5)	0.17
Sex								
Male N[%]	18 [51.4]	17 [63.0]	35 [56.5]	26 [59.1]	0.79	11 [45.8]	15 [75.0]	**0.05**
Female	17 [48.6]	10 [37.0]	27 [43.5]	18 [40.9]		13 [54.2]	5 [25.0]	
Smoking								
Current N[%]*	18 [66.7]	10 [43.5]	28 [56.0]	19 [57.6]	0.79	12 [70.6]	7 [43.8]	0.19
Former	2 [7.4]	8 [34.8]	10 [20.0]	8 [24.2]		2 [11.8]	6 [37.5]	
Never	7 [25.9]	5 [21.7]	12 [24.0]	6 [18.2]		3 [17.7]	3 [18.8]	
Mean BMI (kg m^{-2}) (SD)	29.9 (6.0)	33.1 (9.7)	31.3 (7.9)	31.0 (6.5)	0.86	30.6 (6.5)	31.6 (6.5)	0.62
BMI ≥18.5 – 24.9 (kg m^{-2}) N[%]	8 [22.9]	4 [14.8]	12 [19.4]	8 [18.2]	0.99	5 [20.8]	3 [15.0]	0.88
BMI ≥25.0 – 29.9 (kg m^{-2})	11 [31.4]	7 [25.9]	18 [29.0]	13 [29.6]		7 [29.2]	6 [30.0]	
BMI ≥30 (kg m^{-2})	16 [25.8]	16 [59.3]	32 [51.6]	23 [52.3]		12 [50.0]	11 [55.0]	
Mean waist–hip ratio (SD)	1.0 (0.1)	1.0 (0.1)	1.0 (0.1)	1.0 (0.1)	0.97	1.0 (0.1)	1.0 (0.1)	0.10
Mean HbA1c (mmol mol^{-1}) (SD)	67.5 (23.9)	61.5 (18.8)	65.0 (21.9)	62.0 (21.0)	0.50	66.1 (22.4)	57.3 (18.6)	0.17
Mean Total cholesterol (mmol L^{-1}) (SD)	4.8 (1.1)	4.6 (0.8)	4.7 (1.0)	4.5 (0.9)	0.22	4.5 (1.0)	4.4 (0.9)	0.61
Mean HDL cholesterol (mmol L^{-1}) (SD)	1.0 (0.3)	1.1 (0.2)	1.0 (0.3)	1.0 (0.2)	0.43	1.0 (0.3)	0.9 (0.2)	0.90
Mean hsCRP (mg L^{-1}) (SD)	6.1 (6.3)	6.0 (6.6)	6.0 (6.4)	4.3 (3.9)	0.10	4.1 (3.7)	4.5 (4.2)	0.74
Geometric mean hsCRP (mg L^{-1})†	4.0 [2.8–5.7]	3.7 [2.4–5.6]	3.9 [3.0–5.0]	2.8 [2.1–3.9]	0.14	2.6 [1.6–4.2]	3.1 [2.0–4.9]	0.58
Mean IL-6 (pg mL^{-1}) (SD)	2.9 (2.7)	2.9 (2.5)	2.9 (2.6)	2.9 (2.4)	0.98	3.7 (2.1)	1.8 (2.5)	**0.02**
Mean number of teeth (SD)	23.3 (7.8)	26.0 (5.4)	24.4 (6.9)	23.9 (7.3)	0.73	22.1 (7.9)	26.1 (5.9)	0.07
CDC/AAP case								
Moderate N[%]	20 [57.1]	18 [66.7]	38 [61.3]	–	–	–	–	–
Severe	15 [42.9]	9 [33.3]	24 [38.7]	–	–	–	–	–
Mean gingival bleeding (SD)	1.6 (0.7)	1.6 (0.7)	1.6 (0.7)	1.2 (0.6)	**0.01**	1.0 (0.5)	1.5 (0.6)	**0.01**
Mean extent CAL ≥3 mm % (SD)	64.3 (26.0)	61.8 (25.1)	63.2 (25.4)	54.4 (22.9)	0.07	51.0 (24.9)	58.4 (20.0)	0.29
Mean extent PPD ≥4 mm % (SD)	19.3 (16.1)	18.4 (17.7)	18.9 (16.7)	11.4 (10.8)	**0.01**	10.0 (10.7)	13.0 (11.0)	0.35

Bold *P*-values denote significant differences between the preceding two columns.
Continuous variables presented as mean (SD) unless otherwise stated; Categorical values presented as N column [%].
Whole-group comparison between baseline and 3-month follow-up via independent *t*-test and chi-square. Comparison of randomization groups at 3-month follow-up via independent *t*-test and chi-square.
*Complete smoking data available for 50 participants at baseline and 33 participants at 3-month follow-up. Gingival bleeding: modified from Löe & Silness 'Gingival Index' scoring system (number of teeth with BOP/number of teeth periodontally assessed); BMI: body mass index; HbA1c: Glycated haemoglobin; HDL: high-density lipoprotein; hsCRP: high-sensitivity C-reactive protein; IL-6: interleukin-6.
†Log-transformed geometric mean and [95% confidence interval] presented to correct for the skewed distribution.

Of the 62 participants with diabetes, complete data were available for 44 people that were followed up at the 3-month recall (Fig. 1). In the treatment group, 10 participants were lost to follow-up and one person withdrew between baseline and follow-up leaving 24 participants with complete data to be re-assessed and a loss to follow-up rate of 31.5%. In the control group, six participants were lost to follow-up and one moved interstate leaving 20 participants with complete data, the loss to follow-up rate being 25.9%. Study participants in the treatment group who completed the 3-month follow-up were, on average, 10 years older than those lost to follow-up and had a significantly lower gingival bleeding index (Table 2). In contrast, participants in the control group who completed the 3-month follow-up were more likely to be male and some 8 years younger than those lost to follow-up: mean age (SD) 44.3 years (9.63) versus 52.4 (2.6), $P = 0.002$ respectively.

The single session of non-surgical periodontal therapy provided to participants in the treatment group resulted in small, but not significant reductions in the extent of periodontal pocketing ≥4 mm, ANCOVA difference in means (95% CI) −1.24 (−5.55, 3.06) and gingival bleeding ANCOVA difference in means (95% CI) −0.09 (−0.38, 0.20) (Table 3). There was a short-term increase in HbA1c among participants randomized to the intervention group, ANCOVA difference in means (95% CI) 0.22 (−6.25, 6.69), which was also not statistically significant (Table 3). Post hoc calculations indicate that, with a total sample size of 62 participants with diabetes, the present analysis had a power of 19% with two-sided $P < 0.05$ and that a total of 398 participants equally randomized would have been required to test this hypothesis with sufficient power.

Discussion

In this randomized controlled trial of Aboriginal Australians with periodontitis, non-surgical periodontal therapy did not significantly modify HbA1c in study participants with type 2 diabetes. The present study has arisen from secondary analysis of randomized trial data focusing on vascular endpoints and was not designed to specifically examine changes in glycaemic control. For this reason, post hoc power calculations were conducted.

Given that the intervention provided non-surgical periodontal therapy immediately to those randomized in the treatment group versus delayed intervention for the control, it was not possible to blind study participants from allocation. That noted, the effects of allocation bias on outcomes of periodontal intervention trials are likely to be minimal. Most important in terms of blinding was that the two clinicians involved in clinically examining study participants for inclusion into the study were also the same clinicians that provided the intervention and re-assessed study participants at follow-up. Concerted attempts were made to reduce the effects of detection bias by ensuring that the clinician who initially examined and provided the periodontal therapy to the study participant was not the same clinician that was involved in the re-assessment at follow-up.

Randomized trials may experience attrition of a small proportion of their sample and often this does not influence the trial outcomes (19). The present study collectively followed up 71% of the baseline sample. The internal validity of trial results can be influenced when the characteristics of those

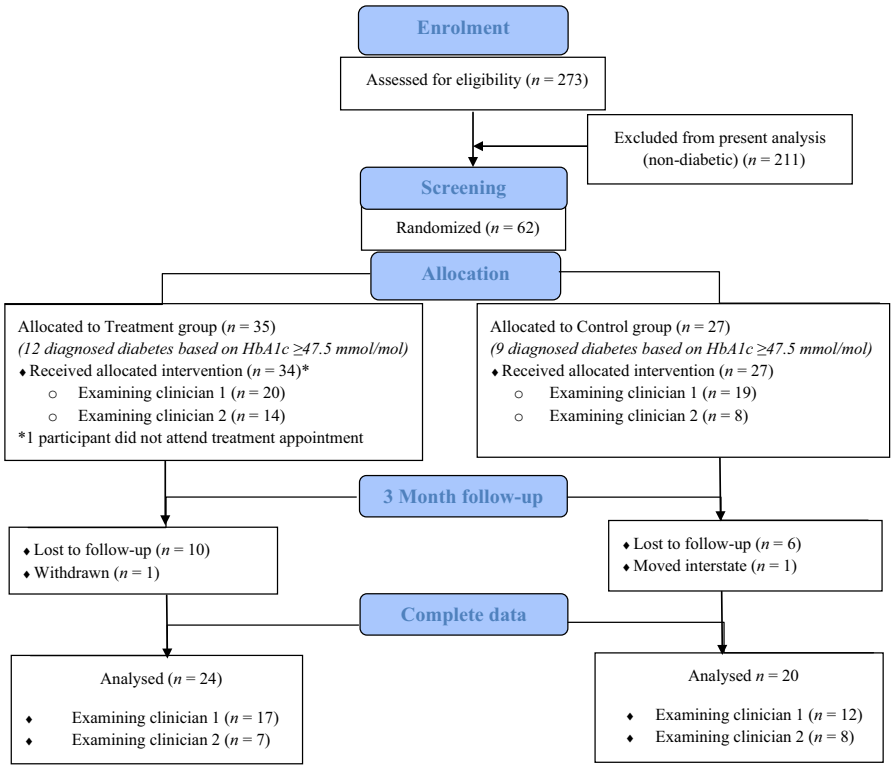

Fig. 1. Study participation.

Table 2. **Baseline characteristics of completed versus lost to follow-up**

Characteristics	Treatment (n = 35)			Control (n = 27)		
	Completed (n = 24)	Lost to follow-up (n = 11)	P-value	Completed (n = 20)	Lost to follow-up (n = 7)	P-value
Mean age (SD)	48.5 (10.7)	38.8 (8.6)	**0.01**	44.3 (9.63)	52.4 (2.6)	**0.002**
Sex						
Male N[%]	11 [45.8]	7 [63.6]	0.33	15 [75.0]	2 [28.6]	**0.03**
Female	13 [54.2]	4 [36.4]		5 [25.0]	5 [71.4]	
Smoking						
Current N[%]*	12 [70.6]	6 [60.0]	0.29	7 [43.8]	3 [42.9]	0.85
Former	2 [11.8]	0 [0]		6 [37.5]	2 [28.6]	
Never	3 [17.7]	4 [40.0]		3 [18.8]	2 [28.6]	
Mean BMI (kg m^{-2}) (SD)	30.4 (6.5)	29.0 (4.9)	0.53	31.7 (7.1)	36.9 (15.0)	0.41
BMI ≥18.5 – 24.9 (kg m^{-2}) N[%]	5 [20.8]	3 [27.3]	0.75	3 [15.0]	1 [14.3]	0.98
BMI ≥25.0 – 29.9 (kg m^{-2})	7 [29.2]	4 [36.4]		5 [25.0]	2 [28.6]	
BMI ≥30 (kg m^{-2})	12 [50.0]	4 [36.4]		12 [60.0]	4 [57.1]	
Mean waist–hip ratio (SD)	1.0 (0.1)	1.0 (0.1)	0.90	1.0 (0.1)	0.9 (0.1)	0.07
Mean HbA1c (mmol mol^{-1}) (SD)	70.3 (25.0)	60.2 (19.9)	0.29	60.8 (20.3)	64.8 (9.0)	0.71
Mean Total cholesterol (mmol L^{-1}) (SD)	4.8 (1.2)	4.9 (1.0)	0.86	4.4 (0.8)	5.2 (0.9)	0.09
Mean HDL cholesterol (mmol L^{-1}) (SD)	1.0 (3.2)	0.8 (0.2)	0.07	1.0 (0.2)	1.2 (0.2)	0.10
Mean hsCRP (mg L^{-1}) (SD)	6.1 (6.8)	6.1 (4.8)	0.98	6.1 (7.0)	5.5 (5.8)	0.84
Geometric mean hsCRP (mg L^{-1})†	3.8 [2.4 – 5.9]	4.5 [2.2 – 9.3]	0.65	3.6 [2.1 – 6.2]	4.0 [1.7 – 9.1]	0.86
Mean IL-6 (pg mL^{-1}) (SD)	3.3 (2.9)	1.9 (1.8)	0.19	2.8 (2.6)	2.9 (2.3)	0.93
Mean number of teeth (SD)	22.5 (8.1)	24.7 (7.1)	0.44	26.2 (25.4)	25.4 (3.7)	0.75
CDC/AAP case						
Moderate N[%]	13 [54.2]	7 [63.6]	0.60	14 [70.0]	4 [57.1]	0.53
Severe	11 [45.8]	4 [36.4]		6 [30.0]	3 [42.9]	
Mean gingival bleeding (SD)	1.4 (0.67)	1.9 (0.6)	**0.03**	1.7 (0.6)	1.3 (0.9)	0.32
Mean extent CAL ≥3 mm % (SD)	60.6 (27.0)	72.4 (22.7)	0.22	64.6 (21.9)	53.7 (33.4)	0.33
Mean extent PPD ≥4 mm % (SD)	17.3 (16.2)	23.6 (15.7)	0.29	18.0 (16.8)	19.8 (21.7)	0.82

Bold P-values denote significant differences between the preceding two columns.
Continuous variables presented as mean (SD) unless otherwise stated; Categorical values presented as N column [%].
*Complete smoking data available for 50 participants at baseline and 33 participants at 3 month follow-up. Gingival bleeding: modified from Löe & Silness 'Gingival Index' scoring system (number of teeth with BOP/number of teeth periodontally assessed); BMI: body mass index; HbA1c: Glycated haemoglobin; HDL: high-density lipoprotein; hsCRP: high-sensitivity C-reactive protein; IL-6: interleukin-6.
†Log-transformed geometric mean and [95% confidence interval] presented to correct for the skewed distribution.

Table 3. **Effects of non-surgical periodontal treatment on HbA1c, hsCRP and clinical periodontal status after 3 months by treatment group (N = 44)**

	Treatment group (n = 24)		Control group (n = 20)		
	Baseline	3 month	Baseline	3 month	ANCOVA LS mean Δ (95% CI)
HbA1c (mmol mol^{-1})	70.3 (25.0)	66.1 (22.4)	60.8 (20.3)	57.3 (18.6)	0.22 (−6.25, 6.69)
hsCRP (mg L^{-1})	6.1 (6.8)	4.1 (3.7)	6.1 (7.0)	4.5 (4.2)	0.64 (−1.08, 2.37)
Mean gingival bleeding	1.4 (0.7)	1.0 (0.5)	1.7 (0.6)	1.5 (0.6)	−0.09 (−0.38, 0.20)
Extent PPD ≥4 mm (%)	17.3 (16.2)	10.0 (10.7)	18.0 (16.8)	13.0 (11.0)	−1.24 (−5.55, 3.06)

Data for baseline/follow-up means presented as mean (SD). Gingival bleeding: modified from Löe & Silness 'Gingival Index' scoring system (number of teeth with BOP/number of teeth periodontally assessed).

completed significantly differ to those lost to follow-up (20). Although the age of study participants within the respective treatment and control groups differed (Table 2), the age of those whom completed the 3-month follow-up was similar (Table 1) between groups to the extent that it is unlikely to impose on the overall study findings. Similarly, the significant difference in sex upon follow-up in the control group (Table 2) is viewed as coincidental in the light of the uneven allocation during baseline randomization which resulted in a 63% male and 37% female distribution (column 3, Table 1). A larger proportion of males were included in the PerioCardio study as a result of recruiting a third of the overall sample from correctional facilities (21). The prison population in the Northern Territory is almost exclusively male, and their higher follow-up rate exacerbated the skewed gender distribution in the control group.

There are several reasons which may explain why the present investigation did not find a significant improvement in HbA1c. Statistical power aside, the overall effect of the intervention on the periodontal status of study participants was small. Although the periodontal therapy was provided during a single session, it was intensive from the perspective that

treatment was continued until all teeth were free from debris. This involved treatment sessions ranging from 45 min up to 3 h. Maintenance of oral hygiene during the study was not reinforced, and consequently, gingival bleeding levels remained consistent with baseline measures after accounting for regression to the mean in the ANCOVA models. Comparisons of the diabetes subsample with the whole PerioCardio study population show that the effect size for short-term reduction in gingival bleeding and 4-mm periodontal pocketing was smaller among participants with diabetes yet consistent based on overlapping 95% confidence intervals (14). Current understanding of the mechanism by which periodontal disease impacts on diabetes is via inflammation which increases insulin resistance. The aim of successful periodontal treatment and oral hygiene instruction is to remove infectious agents and reduce inflammation to the periodontal tissues. Subsequent to regular home oral hygiene practices, glycaemic control is expected to improve among individuals with type 2 diabetes. It is feasible to assume that residual periodontal pocketing and its concomitant inflammation perpetuated chronic inflammation systemically.

Another, more complex interaction that warrants discussion is the circular relationship between adiposity, systemic inflammation and glycaemic control. Recently, this phenomena has been termed "meta-inflammation" to describe the chronic inflammatory state that develops in response to excess nutrient intake and the consequent insulin resistance (22). Findings from the Atherosclerosis Risk in Communities Study highlighted that among lean people, CRP levels were over two times higher in those with severe periodontitis compared to individuals with mild periodontal disease (23). Interestingly, this relationship diminished linearly and periodontal disease severity was not a significant predictor of CRP in obese people with a BMI of $\geq$35 kg m^{-2}. Participants in the PerioCardio study had an average BMI of 31 kg m^{-2}. Therefore, the effect of periodontal inflammation in these clinically obese individuals may be marginal when considering the overall inflammatory burden that adiposity contributes systemically.

A recently published trial (13) and the discussion surrounding its findings has provided some important insights (12). The Diabetes and Periodontal Therapy Trial (DPTT) enrolled 514 participants with both diabetes and periodontal disease in five centres throughout the USA. Following randomization, non-surgical periodontal therapy was provided at baseline along with adjunctive Chlorhexidine mouth rinse. Supportive periodontal maintenance was also provided to the intervention group at 3 and 6 months. From a periodontal status perspective, the intervention, as expected, resulted in significant improvements in periodontal pocketing and bleeding on probing when compared to the untreated control group. However, bleeding on probing persisted in 40% of sites examined at 3 and 6 months post-baseline in the treatment group indicating that, despite three periodontal treatment sessions in the space of 6 months, periodontal inflammation continued.

The DPTT was stopped early by the data safety and monitoring board due to futility as, after 6 months, HbA1c increased in both the treated and untreated groups to the point where it was likely that there would be no benefit to individuals receiving periodontal treatment (13). Concerns have been raised that the study included a majority (~60%) of individuals with HbA1c <8% (63.9 mmol mol^{-1}) with the implication that most of the study participants were already close to the target of <7% (53.0 mmol mol^{-1}) and may not benefit from receiving periodontal treatment to the same extent as somebody with higher baseline HbA1c values (12). Whether this is the case or not is presently unclear. The lower threshold of HbA1c for inclusion into the DPTT was 7% (53.0 mmol mol^{-1}) while the upper limit was 10% (83.6 mmol mol^{-1}). Twenty-two of 514 study participants had HbA1c <7% (53.0 mmol mol^{-1}), technically a study violation (13). As a comparison, participants in the PerioCardio study with diabetes had a mean HbA1c of 65 mmol mol^{-1}, (equivalent to 8.1%) and a range of 37.7—128.4 mmol mol^{-1} (5.6%—13.9%) (data not shown). Almost 60% of participants in the present study had an HbA1c $\geq$53 mmol mol^{-1} at baseline ($\geq$7%) (data not shown), indicative of suboptimal glycaemic control (24). Non-significant findings in the DPTT may also have been due to high adiposity among participants, with a mean BMI over 34 kg m^{-2}.

There has been a considerable interest for some time within the periodontal research community to determine whether treatment of periodontal disease influences glycaemic control in people with diabetes. Over the preceding two decades, numerous intervention studies have been conducted, the results of which are now summarized in systematic reviews and meta-analyses (9, 10, 25). A Cochrane review, published in 2015, pooled 14 randomized trials into meta-analysis and subsequently reported that the average percentage reduction in HbA1c following scaling/root planing and oral hygiene (+/− antibiotic therapy) was −0.29% (95% CI −0.48 to −0.10) (10). Another systematic review which pooled nine randomized trials into meta-analysis reported a similar magnitude reduction in HbA1c −0.27% (95% CI −0.46 to −0.07) (9). It must be noted that the unit of HbA1c measurement in the present study is different to that of the referenced systematic reviews. This modification is in-line with recommendations from the International Federation of Clinical Chemistry which since 2011 has recommended reporting HbA1c in mmol mol^{-1} (26). As a comparison, a 0.22 mmol mol^{-1} change in HbA1c reported in Table 3 is equivalent to a change of one-fifth of 0.1% using the old scale. Collectively, the absence of any notable change in glycaemic control in response to a single untimed session of non-surgical periodontal therapy leads us to conclude that this intervention yielded no tangible benefit in this population.

Emerging evidence from a meta-analysis which included five studies and 42,198 participants suggests that prospective weight gain is associated with periodontitis incidence (27). Insights connecting periodontitis and body weight have arisen from secondary analysis of data which included 55 obese individuals with generalized severe periodontitis but who were otherwise systemically healthy (such as non-diabetic) (28).

When comparisons of these obese individuals were made with the 112 with equivalent periodontitis experience but a normal BMI, periodontal pocketing outcomes were poorer following non-surgical periodontal therapy such that obese individuals had a mean 3% more sites with residual pocketing of 4 mm or more (28). The authors went further to describe the association between periodontal treatment responses in obese individuals as being tantamount to the effects that tobacco smoking has on the periodontium. In a more recent case–control study, being overweight: odds ratio 2.6 (95% CI 1.2 – 5.4) or obese: OR 3.1 (95% CI 1.1 – 6.5) was associated with up to a threefold risk of having periodontitis compared to individuals with a BMI below 25 kg m^{-2} (29). In this population, BMI was a stronger risk factor than current smoking OR 2.5 (95% CI 1.1 – 5.6) or former smoking OR 2.6 (95% CI 1.2 – 5.9) (29).

Dietary intake, particularly of carbohydrates, is a key determinant of glycaemic control in type 2 diabetes and obesity (30). Participants in the present study were involved in the PerioCardio study which aimed to investigate the relationship between periodontitis, arterial stiffness and function. Information concerning diet was not collected, and consequently, a definitive determination on whether carbohydrate intake may have influenced the present study findings in relation to HbA1c and periodontal therapy response is not possible. Nevertheless, the study team did not discuss the potential effects of diet beyond its influence on dental caries at any time during the trial and thus speculate that diet would have remained unchanged.

The practical implication for dental clinicians is that poor management of glycaemia over a prolonged period increases the risk of diabetic complications that include impaired healing (31) and having severe periodontal disease (32). Convincing evidence exists that periodontal treatment is effective in reducing HbA1c by clinically significant levels in among people with a healthy BMI (33–35). Findings from previous trials (13, 36) along with the present suggest that when individuals are obese, periodontal treatment may not influence HbA1c. Based on the few studies (including the present) that have provided some form of periodontal intervention to individuals with diabetes that also have a high BMI, it is precisely unclear whether this lack of improvement in HbA1c was due to adiposity, due to persistent periodontal inflammation or a combination of both. Furthermore, an attenuated periodontal tissue response to periodontal therapy among overweight/obese individuals may suggest that a more intensive initial phase approach is warranted in addition to a recall period shorter than 3 months.

Both periodontitis (37, 38) and obesity (39–41) impact on glycaemic control via inflammation. In the obese state, white adipose tissue stimulates tumour necrosis factor-alpha and other pro-inflammatory cytokines and weight loss is known to attenuate this pattern (42, 43). Additionally, obesity is known to influence glycaemic control directly via dyslipidaemia. A fine review connecting inflammation, hepatic and muscular lipid accumulation with impaired glucose tolerance and type 2 diabetes has been published (44).

This poses a clear direction for future studies but also potential statistical challenges in analysing the data which eventuates. To truly examine the effect of periodontal treatment on glycaemic control among individuals with obesity, periodontal disease must be treated and managed to the point where periodontal infection and inflammation is eliminated. Intervention studies falling short of attaining this clinical endpoint such as the present and previous studies (13, 36) are unable to wholly answer this important question. A shortened recall period of perhaps 4 to 6 weeks as opposed to 3 months may be required to gauge the immediate response while also providing clinicians with the option of instigating rescue therapy to non-responsive sites and reinforcing oral hygiene methods with the patient. This approach to treatment will require considerable clinical resources as well as financial investment, especially with study samples involving hundreds of participants. Completed periodontal treatment will allow for the possibility to conduct analysis of mediation, to quantify the magnitude of glycaemic control disturbance attributed solely to periodontitis. Two potential statistical methods on how this can be achieved are available in the literature (45, 46).

The results from this underpowered study suggest a lack of any measurable effect 3 months after one session of non-surgical periodontal treatment and oral hygiene instructions on glycated haemoglobin levels in overweight/obese individuals. The results concur with other studies reporting that HbA1c may not notably respond to periodontal treatment in individuals with obesity. Future studies in this field should aim to definitively manage the periodontal state in the first instance, if only for its periodontal benefits. Testing the effects of periodontal treatment on HbA1c in obese patients requires additional studies.

Clinical relevance

Scientific rationale

The prevalence of type 2 diabetes and obesity is increasing globally. Consequently, more individuals presenting for periodontal care are likely to have both conditions.

Principal findings

The present study supports findings from other periodontal intervention trials that conventional periodontal therapy does not reduce glycated haemoglobin in individuals with diabetes and obesity.

Practical implications

Given it appears that the tissue healing and anti-inflammatory responses to periodontal therapy are diminished in diabetes individuals with obesity, management may require shorter recalls and more intensive interventions than both non-diabetic and people with diabetes and a healthy BMI.

Acknowledgements

The authors gratefully acknowledges the support of the PerioCardio participants, study staff, Danila Dilba Health services, Northern Territory Oral Health Services, Northern Territory Correctional Services, the Baker IDI Heart and Diabetes Institute Alice Springs, the Wurli Wurlinjang Aboriginal Health Service and the South Australian Dental Service who kindly donated a dental chair to the Wurli Wurlinjang Aboriginal Health Service. The PerioCardio study was funded by the National Health and Medical Research Council: Project grant #627100. MRS is supported by a Future Leader Fellowship from the National Heart Foundation of Australia #100419. KK received a University of Adelaide Divisional Scholarship to participate in this research.

Conflict of interest

The authors report no conflict of interest in the production or publication of this work.

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